Oct 15, 2026 Version 2

Islet Culture and Preparation for Cold Shipping V.2

DOI
https://dx.doi.org/10.17504/protocols.io.4r3l28poql1y/v2

Integrated Islet Distribution Program[1]

[1][Integrated Islet Distribution Program, City of Hope, Duarte, CA]

Integrated Islet Distributi... Barb's interest 1 more workspace

Integrated Islet Distribution Program
Integrated Islet Distribution Program, City of Hope

OPEN ACCESS

DOI: https://dx.doi.org/10.17504/protocols.io.4r3l28poql1y/v2

External link: https://iidp.coh.org/Investigators/Policies-Standard-Operating-Procedures

Protocol Citation:
Integrated Islet Distribution Program 2026. Islet Culture and Preparation for Cold Shipping. **protocols.io**
https://dx.doi.org/10.17504/protocols.io.4r3l28poql1y/v2
Version created by
Barbara Olack

License: This is an open access protocol distributed under the terms of the **Creative Commons Attribution License**, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited

Protocol status: Working
We use this protocol and it's working

Created: January 25, 2022

Last Modified: October 15, 2026

Protocol Integer ID: 57414

Keywords: Human islets, Dithizone, Purity, Islet Equivalent, Actual Islet, Islet Quality Grade, Islet Index, research in the integrated islet distribution program, integrated islet distribution program, culture of the islets post isolation, islets post isolation, recovery of the isolated islet, islets by the prodo lab, participating islet isolation center, isolated islet, islet isolation center, iidp islet, concentrations of islet, islet culture, islet, preferred islet media, post isolation preparation, preparation for cold shipping, allocation of the post isolation preparation, national institute of diabetes, diabetes, iidp sop, cold shipping, culture flask, isolation process, original iidp method,

Funders Acknowledgements:
NIDDK
Grant ID: U24DK098085

Abstract

This SOP describes the culture of the islets post isolation, to allow the recovery of the isolated islets from the trauma of the isolation process. This includes the allocation of the post isolation preparation into defined concentrations of islets per culture flask, using standardized, preferred Islet Media for 37°C culture. This SOP also provides instructions for a half media change of all culture flasks, within 18-24 hours of original culture, following isolation. Islets are kept on culture for a minimum of 18-24 hours prior to shipment for use in the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) sponsored research in the Integrated Islet Distribution Program (IIDP). This protocol is written to assist the participating islet isolation centers and investigators who are part of this program.

Note

Integrated Islet Distribution Program (IIDP) (RRID:SCR _014387)

Note: This SOP was developed based on the Prodo Labs, Inc. shipping protocol and results from preliminary studies conducted by the IIDP and commissioned by the original IIDP Project Officer, and External Evaluation Committee (EEC).

It was commissioned due to problems with acquiring the supplies that were used in IIDP SOP: SHP-001 and with the hope that this method is a better means for transportation of IIDP islets. Preliminary studies proved the islets by the Prodo Labs' method were statistically as good as the original IIDP method, it was preferred by the test researchers, and was much more cost effective for the IIDP. This new method may be modified as future methods are tested and approved by the IIDP Team, Project Scientist (PS), the Program Official (PO), and EEC.

Guidelines

Responsibilities

- It is the responsibility of the IIDP CC to both follow and ensure adherence to the procedures outlined in this SOP. In order to accomplish this, the IIDP CC will interact with the relevant personnel from each of the participating centers.

- It is the responsibility of each IIDP center to follow the procedures listed in this SOP and to work to the best of their ability to follow all requirements.

Definitions

- ***Integrated Islet Distribution Program (IIDP) (RRID:SCR_014387):*** The IIDP is a contracted program commissioned and funded by the NIDDK to provide quality human islets to the diabetes research community to advance scientific discoveries and translational medicine. The IIDP consists of the NIDDK, the Project Officer (PO), the External Evaluation Committee (EEC) and the Coordinating Center (CC) at City of Hope (COH). The IIDP CC integrates an interactive group of academic laboratories including the subcontracted IIDP centers.

- ***IIDP Coordinating Center (CC):*** Joyce Niland, Ph.D. is the Principal Investigator for the IIDP CC and leads staff from the Department of Diabetes and Cancer Discovery Science, Diabetes and Metabolism Research Institute at COH to coordinate the activities of the IIDP and assist the participating centers and investigators in the distribution of human islets.

- ***Islet Equivalent (IEQ)***: A conversion factor of the size of an actual islet to an equivalent size of an islet which is 150 µm diameter by mathematically compensating for the volume.

- ***Approved Investigators***: Researchers who have requested islets from the IIDP for basic science studies and whose research protocols have been reviewed and approved by the IIDP.

- ***Islet Allocation System (IA)***: This is the online system administered by the IIDP to allow fair distribution of basic science islets to approved investigators. This interactive system is used by the IIDP Centers and the approved investigators facilitates and tracks the distribution of human islets.

Materials

MATERIALS

- Dithizone (Diphenylthiocarbazone) **Merck MilliporeSigma (Sigma-Aldrich) Catalog #**D5130
- Gibco DPBS without Calcium and Magnesium **Fisher Scientific Catalog #**14190136 or equivalent
- Dimethyl sulfoxide **Merck MilliporeSigma (Sigma-Aldrich) Catalog #**D8779 or equivalent
- Human AB Serum (ABS) HI **Gemini Bio-Products Catalog #**100-512; Heat Inactivated
- PIM(G)® (5 mL Glutamine/Glutathione) **Prodo Laboratories, Inc Catalog #**PIM(G)®
- Corning™ Ciprofloxacin Hydrochloride **Fisher Scientific Catalog #**MT61277RG (Corning™ 61277RG)
- PIM(R)® **Prodo Laboratories, Inc Catalog #**PIM(R)®
- MP Biomedicals™ Ciprofloxacin Hydrochloride or equivalent **Fisher Scientific Catalog #**MP219902005

Equipment

Gilson™ PIPETMAN Classic™ Pipets- F123601 or equivalent	NAME
Adjustable pipettor -P200	TYPE
Gilson	BRAND
F123601G	SKU
https://www.fishersci.com/shop/products/gilson-pipetman-classic-pipets-model-p200-vol-50-200-l/f123601g	LINK
50 to 200µL, ±0.5, ±1µL	SPECIFICATIONS



Equipment

Fisherbrand™ Large-Orifice Pipet Tips, 1 to 200µL or equivalent	NAME
Genomic/Wide Orifice Pipet Tips	TYPE
Fisherbrand	BRAND
02-707-134	SKU
https://www.fishersci.com/shop/products/fisherbrand-large-orifice-pipet-tips-6/02707134	LINK
1 to 200µL	SPECIFICATIONS



Equipment

Fisherbrand™ Petri Dishes with Clear Lid or equivalent

NAME

Petri Dishes

TYPE

Fisherbrand

BRAND

FB0875713A

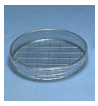
SKU

https://www.fishersci.com/shop/products/fisherbrand-petri-dishes-clear-lid-12/fb0875713a

LINK

Round, Raised Ridge, 60mm, 15mm

SPECIFICATIONS



Equipment

Bottle Top Filters with PES Membrane or equivalent

NAME

0.2 µm Bottle Top Filter

TYPE

Thermo Scientific™ Nalgene™ Rapid-Flow™ Sterile Di

BRAND

09-741-09

SKU

https://www.fishersci.com/shop/products/nalgene-rapid-flow-sterile-disposable-bottle-top-filters-pes-membrane/0974109

LINK

0.2 µm membranes for sterile filtration; PES is the best membrane for cell culture fluids; lowest protein binding to maintain protein balance, lowest extractables to maintain media purity

SPECIFICATIONS



Equipment

Invitrogen™ EVOS™ XL Core Imaging System

NAME

Digital, transmitted light, inverted imaging system or equivalent

TYPE

Invitrogen Microscope

BRAND

12-562-751

SKU

https://www.fishersci.com/shop/products/evos-xl-core-imaging-system/12562751#?keyword=Inverted+Microscopes

LINK

EVOS XL Core Imaging System with fixed stage

SPECIFICATIONS



Equipment

Bal Supply Cell Counter or equivalent

NAME

Cell Counter (Manual or Electronic)

TYPE

Bal Supply

BRAND

02-670-14

SKU

https://www.fishersci.com/shop/products/bal-supply-cell-counters-4/0267014

LINK



Equipment

Drummond™ Fixed-Volume Microdispensers or equivalent

NAME

Drummond 3000385

TYPE

Drummond™

BRAND

21176F

SKU

https://www.fishersci.com/shop/products/drummond-fixed-volume-microdispensers-16/21176f?searchHijack=true&searchTerm=21176F&searchType=RAPID&matchedCatNo=21176F

LINK

Volumetric Range 100/200UL with borosilicate glass bores

SPECIFICATIONS

Equipment

Corning 4492 or equivalent

NAME

Sterile Polystyrene Serological Pipets

TYPE

Corning™ Stripette™ Wide-Tip Disposable Polystyren

BRAND

07-200-619

SKU

https://www.fishersci.com/shop/products/stripette-disposable-plastic-wrapped-polystyrene-serological-pipettes-2/07200619?searchHijack=true&searchTerm=07200619&searchType=RAPID&matchedCatNo=07200619

LINK

Sterile, individually wrapped, and certified nonpyrogenic and DNase-/RNase-free
Accuracy within ±2% at full volume

SPECIFICATIONS



Equipment

Thermo Scientific™ Nunc™ Non-treated Flasks

NAME

T-175 Non-treated Flasks

TYPE

Thermo Scientific™ Nunc™

BRAND

12-566-85

SKU

https://www.fishersci.com/shop/products/nunc-non-treated-flasks/1256685

LINK

T-175 Non-treated Flasks

SPECIFICATIONS



Safety warnings

Please see attached SDS (Safety Data Sheet) for hazards and safety warnings.

Ciprofloxacin Hydrochloride Ciprofloxacin SDS-2014Jul22.pdf
MP Cipro_MSDS_199020_USA_EN....

Precautionary statements:

- P280 - Wear protective gloves and eye/face protection
- P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
- P337 + P313 - If eye irritation persists: Get medical advice/attention.
- P273 - Avoid release to the environment.

GemCell™ U.S. Origin Human Serum AB GemCell Human Serum AB.pdf

GemCell™ human serum AB is collected from healthy male donors of the AB serotype at FDA-licensed facilities in the United States.

Hazardous Components:

- Biohazard contains human source material. Handle as though capable of transmitting infectious agents.
- Toxicity: Not Established.

Target Organs/Systems: Product could possibly irritate the skin, eyes and respiratory system. Do not ingest this product.

Dimethyl sulfoxide (DMSO) DMSO_MSDSAction.pdf

- Hazard statement(s): Combustible liquid.
- Precautionary statement(s): Keep away from heat/sparks/open flames/hot surfaces. No smoking. Wear protective gloves/ protective clothing/ eye protection/ face protection.
- DMSO itself is not toxic but it can be a carrier of chemicals, viruses, etc. into the skin

Dithizone (DTZ) Dithizone_MSDSAction.pdf

- Hazard Statement(s): Causes skin irritation. Causes serious eye irritation.

Precautionary statement(s): Wash skin thoroughly after handling. Wear protective gloves/ eye protection/ face protection.

Ethics statement

Samples used for IIDP protocols are not from living donors and are exempt from IRB approval.

Preparation of Supplies and Reagents

1

Laboratory Supplies:

The following supplies are necessary for the preparation of flasks for human islet culture prior to distribution.

- Islet preparation and distribution
- Wide mouth pipettes and pipettor
- Culture flasks- T-175 non-coated flasks
- Sampling wide-bore pipette tip and pipettor (Gilson or Drummond)
- 37°C CO_2 Incubator
- Conical tubes, 250 ml, sterile
- Routine lab supplies for transferring, media changing and counting islets.

2 **The IIDP will provide each center with the following supplies necessary for islet culture:**

- Human AB Serum (ABS) HI
- PIM(G)®
- PIM(R)®
- Ciprofloxacin Hydrochloride

3 **Receipt of Supplies:**

The majority of supplies should be stored in appropriate dry, temperature-controlled environments (room temperature 16°-28°C).

- The Prodo Labs PIM(R) should be stored, in the dark, between 2 °C and 8 °C upon receipt but is stable at room temperature.
- The Gemini AB serum and the PIM(G) vials should be stored at -5 °C to -20 °C in the dark.
- The Ciprofloxacin should be stored as indicated by manufacturer (Corning at room temperature OR MP Biomedicals at 4°C). Aliquots of Ciprofloxin can be prepared prior to the isolation. Prepare the Ciprofloxacin according to the directions in the table. Filter sterilized suspension aliquots should be stored at -5 °C to -20 °C .

Pre-Preparation of Ciprofloxacin Powder for Addition to Media

- Remove 0.5 gm (500mg) of ciprofloxacin hydrochloride from the bottle and QS to 50mL with distilled water. This will give a stock concentration of 10mg/ml.
- Mix with a stir bar and stirring plate until totally dissolved.
- Filter sterilize the solution using a 0.2µM filter.
- Aliquot into sterile tubes, 5mL samples, label, and freeze for later use.
- The expiration date of the solution is indicated on the Certificate of Analysis and/or the bottle. Document expiration date as date of CoA.
- Diluted solution is good for 1 year frozen (if less than CoA expiration date) and 1 month thawed.

Record Ciprofloxacin preparation on ***Attachment 1: Solutions Preparation Sheet****, of this SOP.*

Attachment 1-Solutions preparation...

4 Preparation of PIM(R) Media:

- Prepare one 500 mL bottle of PIM(R) media prior to the isolation
- Thaw and add 5 mL of PIM(G)
- Thaw and add 25 mL of AB serum (5% v/v)
- Thaw and add 0.5 mL of prepared ciprofloxacin sterile aliquot
- Once all additives have been added to the bottle of PIM(R), it is now referred to as PIM(R) complete.
- Note: If culture for IIDP distribution is greater than 250,000 IEQ, a second bottle of PIM(R) complete will need to be prepared.

Note

If PIM(R) complete is to be used from a previous isolation, it must have been filter sterilized at the end of the previous use. The media will expire within 30 days, once it has been fully supplemented.

Note

Record media preparation on ***Attachment 1: Solutions Preparation Sheet****, of this SOP.*

Post Purification Culture of Islet with ≥70% Purity

5 Post purification, all pooled islets with > 70% purity should be brought up in 200 mL of PIM(R) in a 250 mL conical.

- Thoroughly mix suspension by either pouring between 2 conicals or inverting one conical at least 3 times.
- Quickly removing cap and count samples should be taken by a second technician.
- Take replicate 100-200 µl sample volumes from 100 mL final prep using a sampling wide-bore pipette tip and pipettor (Gilson or Drummond). Two duplicate counts should be performed by two separate technicians or 2 separate counts by one technician.
- Add 3 drops (30 µL) of the DTZ solution to the islets sample and allow staining for 1 – 2 minutes at room temperature. Cover the bottom of the counting dish with DPBS to approximately ½ the height of the dish. Count the islets under the microscope as described in the SOP: ***Qualitative & Quantitative Assessment of Human Islets for Distribution Using Dithizone (DTZ) V.2*** https://www.protocols.io/view/qualitative-quantitative-assessment-of-human-islet-bhdjj24n

Note

Note: This can be accomplished using center specific method as long as a representative sample is taken from a well-mixed suspension and the sample is suspended in PIM(R)

- While counting, place the conical that contains the islets on its side at room temperature in the hood so that the islets do not pellet.

Note

Record counts on Attachment 2-Islet Tabulation Counting Sheet for Culturing Islets using Attachment 3- Islet Ranking Guide for Ranking, of this SOP.

Attachment 2-Islet Tabulation Coun... Attachment 3-Islet Ranking Guide.p...

6 Once the average count has been determined from the samples taken and recorded on the count sheets, calculate the total IEQs and the IEQ per mL in the islet suspension.

- Example: If 250,000 IEQ were in the 200 mL suspension, then 250,000 IEQ/200 mL =1,250 IEQ/mL.

7 Calculate the amount of suspension needed to aliquot 20,000 IEQ per T-175 flask.

- Example: 20,000 IEQ/1,250 IEQ/mL = 16 mL/Flask

8 Calculate the number of flasks needed to culture the islets for broadcast at 20,000 IEQ per flask.

- Example: 250,000 IEQ/20,000 IEQ=12.5 flasks (or 200 mL/16 mL=12.5.) Round up the fractional number of flasks to the next whole number and recalculate the proper aliquot to be taken from the islet suspension.
- Example: 12.5 → 13 Flasks; 200 mL/13 flasks = 15.4 mL islet suspension/flask.
- Determine the amount of media needed to prewet each flask. Example: 40 mL - 15.4 mL islet suspension = 24.6 mL fresh PIM(R).

9 Aseptically transfer the proper amount of T-175 non-coated flasks into the hood and label per center protocol. Pre-wet all but one flask with predetermined amount of fresh PIM(R).

- For the final flask, pre-wet with 6 mL less than the calculated amount and mark as final flask.

10 Lay the flasks flat on the hood surface making sure the media covers the entire surface being careful not to wet the neck or cap of the flask.

11 Deliver ~20,000 IEQ to appropriate flasks with a 25 mL pipette properly mixing the islet containing solution between each flask to ensure even distribution.

12 Rinse the conical after the (marked) final flask has been loaded with 3 mL of extra media. Repeat. Total volume will be 40 mL

13 After dispensing all of the islets, place all the flasks in the incubator and set at 37°C ± 0.5°C and 5% ± 0.5% CO_2 and culture for at least 48 hours.

Note

Note: Possible exceptions for culture time.

Attachment 4- Tips for Determining...

14 Between 12 and 18 hours after culture begins, perform a 50% media change on all flasks listed in Step 16-26 under ***Half Media Change Procedure.***

Post Purification Culture of Islet with <70% Purity

15 Follow all steps outlined in steps 5-14, however determine the concentration of the cultured islets and the number of flasks needed in Step 7 (bullet point) by multiplying the total flasks by the percent purity.

- Example: If part of the prep has a 50% purity and 50,000 IEQ; then 50,000 IEQ/(20,000IEQ * 50%) = 5 flasks.
- Determine all other values based on this calculation.

Note

Note: Islets with < 50% purity are not eligible for reimbursement.

Half Media Change Procedure

16 A media change of half the culture volume should be performed 18 -24 hours after the completion of the isolation, Day 1 post culture. If for some reason the islets cannot be shipped out before or on Day 5, a second media change should be performed. (*Note: It is unlikely that this scenario will occur.)*

17 Warm up required amount of PIM(R) to room temperature by taking it out of the refrigerator at least 1 hour before the media change is performed. Aseptically place into the laminar flow hood. 00:00:00

18 Remove the flask(s) from the incubator, keeping them vertical (with the cap facing upward) while transporting from incubator and aseptically place the flask(s) into laminar flow hood.

19 Rock the flasks gently back and forth keeping them horizontal, to get any islets that are loosely attached to the bottom of the flasks to go into the solution, being careful avoiding any media getting into the caps of the flasks. Return upright and loosen the caps of the flask(s).

20 Arrange a 50 mL conical rack supported by a 250 mL conical rack for each set of 10 flasks. Placing 5 in each row, position flask(s) at an angle, tilting it towards the longer

edge using 250 mL and 50 mL conical racks for support, resting them on the longer edge of the 50 mL conical rack so all islets can congregate to the lower corner of the flask due to gravity. Start a timer set for 45 minutes and record start time in the batch record.
00:00:00

21 Leave all flask(s) positioned in this manner for 45 minutes to allow islets to settle in the bottom corner surface.

22 Go to the middle point of the media in one flask and take a 1 mL sample, place in a Petri dish, and examine under the microscope. If islets are less than 50 micron in size proceed to the next step. If larger islets are visible allow more settling time until the media sample has no islets present.

23 While maintaining the position of the flask (as in step 19) aspirate 50% of the used media (20 mL) from the surface of the liquid layer equidistant from the sides of the flask, without disturbing the islets that are settled in the bottom corner.

24 Pipette the aspirated used media into an empty sterile container. (This can be checked when the media change is completed for viable islets that can then be returned to a flask.) Repeat steps for all remaining flasks.

25 Pipette 20 mL of fresh PIM(R) into each flask after aspiration and tighten caps.

26 Once media change is completed for all the flask(s), check aspirate for any greater than 50 µm islets and if any are found, transfer to a flask. Confirm all ventilated flask caps are tight and return to incubator set to 37°± 0.5°C.

27 *Using a 0.2 µm filter, filter sterilize any media remaining in the fresh PIM(R) complete bottle. The media will expire within 30 days once it has been fully supplemented.*

Protocol references

Scharp DW, Arulmoli J, Morgan K, Sunshine H, Hao E: **Advances in Human Islet Processing: Manufacturing Steps to Achieve Predictable Islet Outcomes from Research Pancreases**. OBM Transplantation 2019 Feb, Volume 3, Issue 1(Current Advancement of Islet Cell Transplantation in the Treatment of Diabetes Mellitus).

Ricordi C, Gray DW, Hering BJ, Kaufman DB, Warnock GL, Kneteman NM, Lake SP, London NJ, Socci C, Alejandro R, et al.: **Islet isolation assessment in man and large animals**. Acta Diabetol Lat 1990 Jul-Sep, 27(3):185-195.

Prodo Labs, Inc. Protocols and Website: http://www.prodolabs.com

Acknowledgements

The Integrated Islet Distribution Program (IIDP)Staff at City of Hope who helped to establish this protocol and the Staff of the IIDP Islet Isolation Centers who use this protocol for standardization of the IIDP.